

CompTIA.

A+ Core 1

Exam 220-1201

A+ 220 1201

Module 6



Configuring Network Addressing and Internet Connections

Prepare for A+ Core 1 by:

- Comparing Internet connection types
- Using basic TCP/IP concepts
- Comparing protocols and ports
- Comparing network configuration concepts

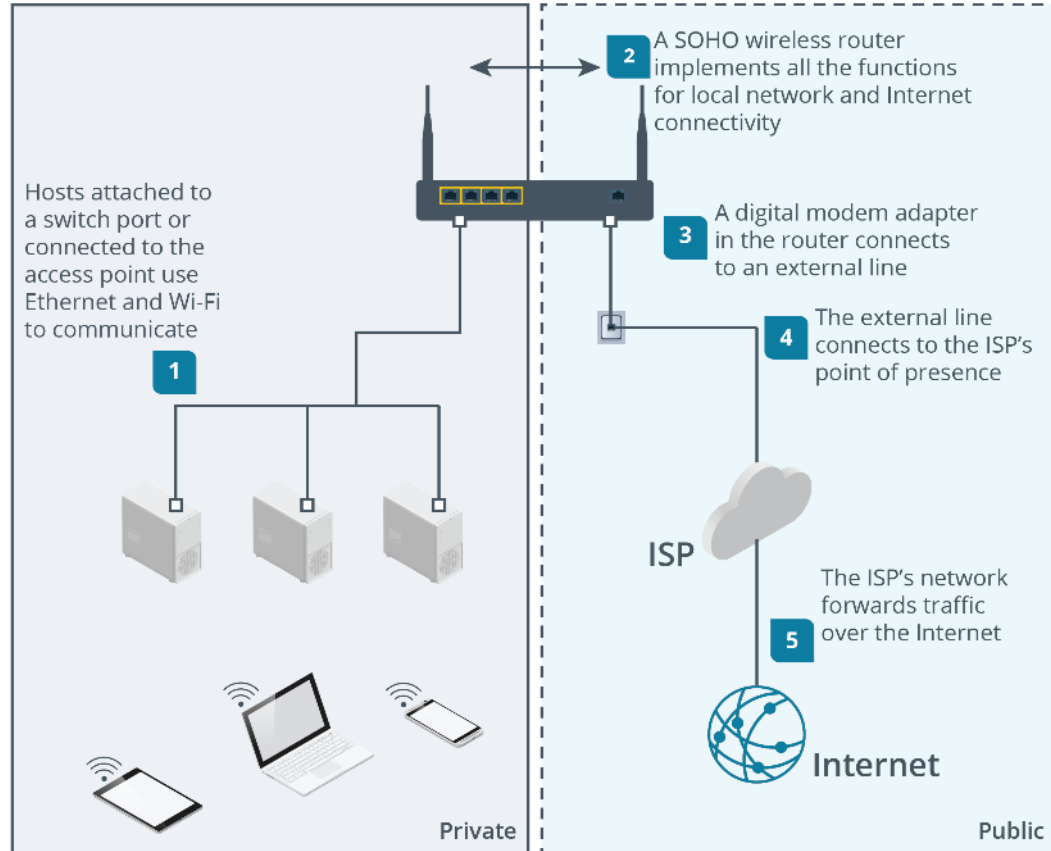
Lesson 6.1

Internet Connection Types

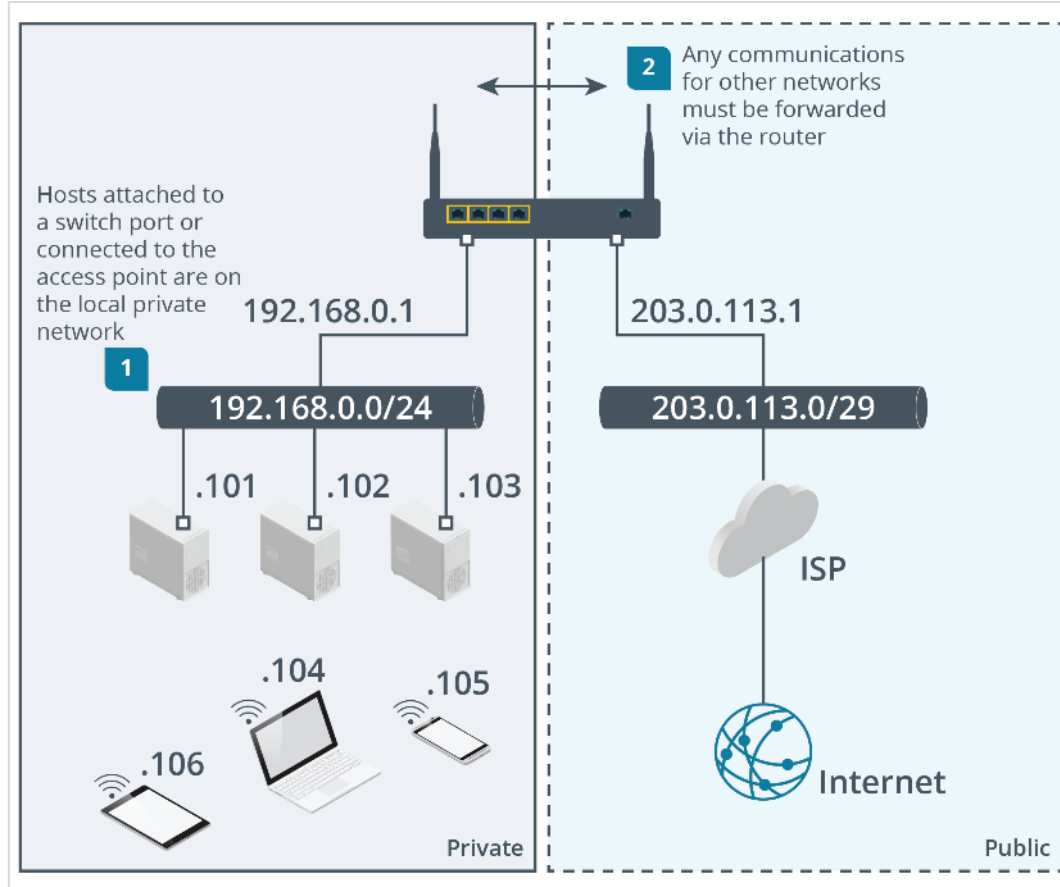
Internet Connection Types and Modems

- **IXP** - **Internet Exchange Point**, is a physical location where different Internet networks (like ISPs, content providers, cloud companies) connect to directly exchange traffic (data) with each other.
- **ISP** - **Internet Service Provider**, a company that gives individuals and organizations access to the internet, acting as the pipeline for online data.
- **Modem**- Establish connection between router WAN interface and ISP

Internet Connection Types and Modems



Internet Connection Types and Modems



Digital Subscriber Line Modems

- Public switch telephone network (PSTN)
- Copper two-pair cabling



Image © 123RF.com.

Digital Subscriber Line Modems

- Asymmetric- Faster download than upload
- Symmetric- Same upload and download speeds
- Filter/Splitter- Voice and data

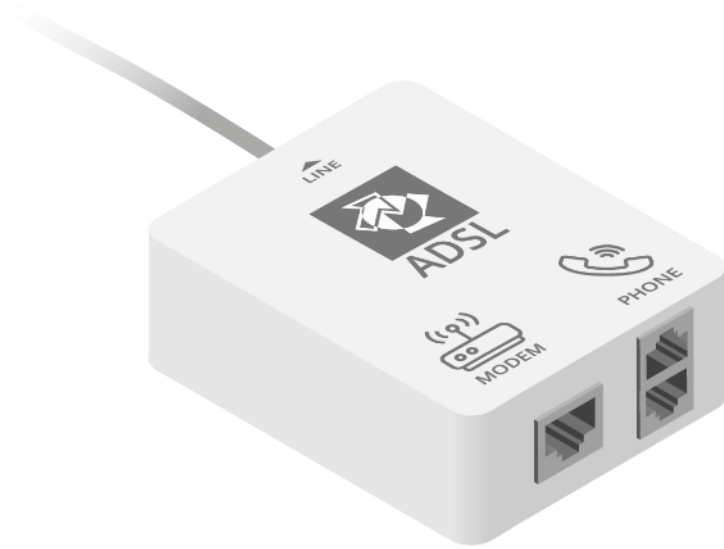


Image © 123RF.com.

Cable Modems

- Cable TV network
- Coaxial cable with F-type connector
- **DOCSIS** – DOCSIS (Data Over Cable Service Interface Specification), a technology standard for high-speed internet over cable networks



Image © 123RF.com.

Fiber to the Curb and Fiber to the Premises



Image by artush © 123RF.com

- FTTC- copper cabling into facility or home
- FTTP- full fiber cable into facility or home
- Optical Network Terminal- ONT

Fixed Wireless Internet Access

Satellite- High latency connection

- Geostationary orbital satellite
- Low Earth Orbital (LEO) satellite

Wireless Internet Service Provider (WISP)

- Long range Wi-Fi connectivity

Cellular Radio Internet Connections

3G

- GSM - (Global System for Mobile Communications)
- CDMA - (Code Division Multiple Access)

4G

- LTE support GSM and CDMA

5G

- mMIMO - (Massive Multiple Input/Multiple Output) is a key 5G wireless technology

Routers

- IP address-based forwarding
- Broadcast domains



Image © 123RF.com.

Firewalls

- ACL- List of rules to allow/deny traffic
- Based on packet header information
 - IP
 - MAC
 - Protocol
 - Port

Firewalls

Firewall: Rules: WAN

Select category

Inspect

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description ?	+ ← ☐
								Automatically generated rules	▼ 4
✖ → ⚡ i	IPv4 *	<bogons>	*	*	*	*	*	Block bogon IPv4 networks from WAN	🔍
✖ → ⚡ i	IPv6 *	<bogonsv6>	*	*	*	*	*	Block bogon IPv6 networks from WAN	🔍
✖ → ⚡ i	IPv4 *	10.0.0.0/8,127.0.0.0/8,100.64.0.0/10,172.16.0.0/12,192.168.0.0/16	*	*	*	*	*	Block private networks from WAN	🔍
✖ → ⚡ i	IPv6 *	fc00::/7	*	*	*	*	*	Block private networks from WAN	🔍
☐	→ ⚡ i	IPv4 ICMP	*	This Firewall	*	*	*	Allow ping to firewall interface	←
☐	→ ⚡ i	IPv4 TCP	*	SCREENED net	80 (HTTP)	*	*	Allow web access (unencrypted)	←
☐	→ ⚡ i	IPv4 TCP	MAILHOSTS	SCREENED net	25 (SMTP)	*	*	Allow SMTP access from secure mail gateway	←
pass pass (disabled)	✖ block ✖ block (disabled)	✖ reject ✖ reject (disabled)	i log i log (disabled)	→ in ← out	⚡ first match ⚡ last match				
Active/Inactive Schedule (click to view/edit)									
Alias (click to view/edit)									

Screenshot used with permission from OPNsense.

Lab Activity

- Lab: Connect a Cable Modem
(available in CertMaster Learn and CertMaster Perform)
- Lab: Connect Fiber Optic Cables
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Internet Connection Types

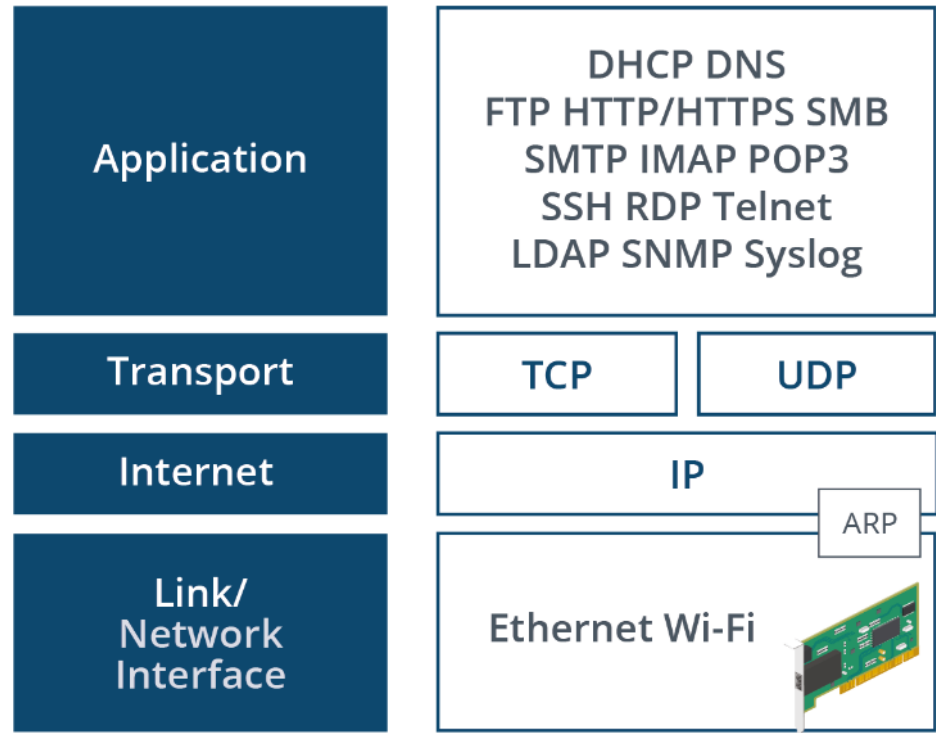
- Internet Connection Types and Modems
- Digital Subscriber Line Modems
- Cable Modems
- Fiber to the Curb and Fiber to the Premises
- Fixed Wireless Internet Access
- Cellular Radio Internet Connections
- Routers
- Firewalls

Lesson 6.2

TCP/IP Concepts

TCP/IP

- Suite of communication protocols



Link or Network Interface Layer

Local network segment
communications

MAC address used for
forwarding

PDU- Frame

Internet Layer

Communication between network segments

IP Address used for forwarding

PDU- Packet

Lesson Review: Transport Layer



TCP VS UDP PROTOCOLS



PDU- SEGMENT

Application Layer



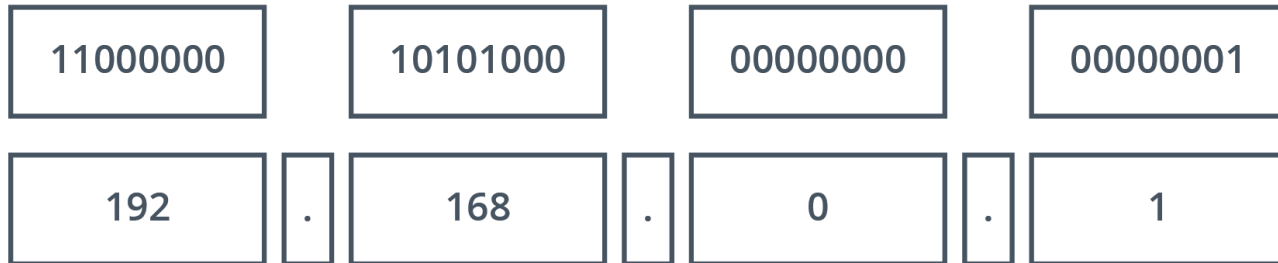
High level functionality
for communications



PDU- Data

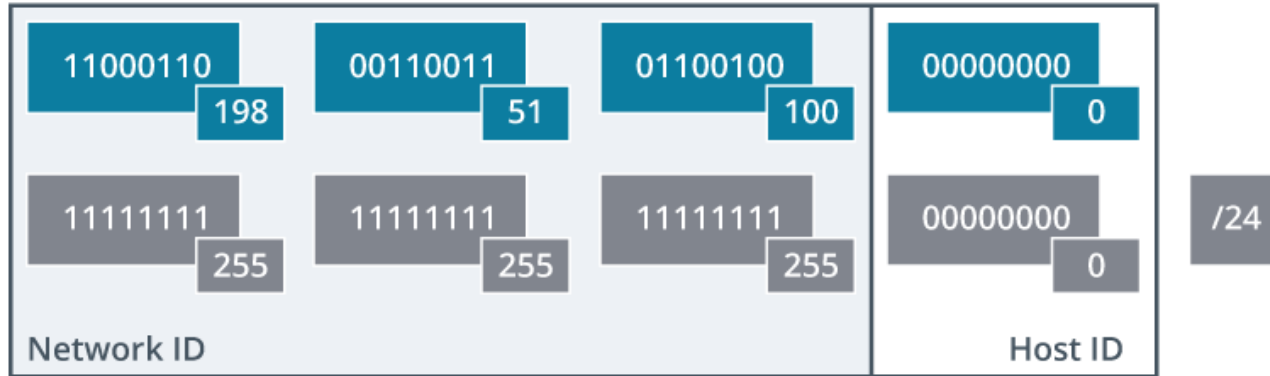
IPv4 Addressing

- 32-bit, dotted decimal notation
- 4 octets
- 0-255 in each octet



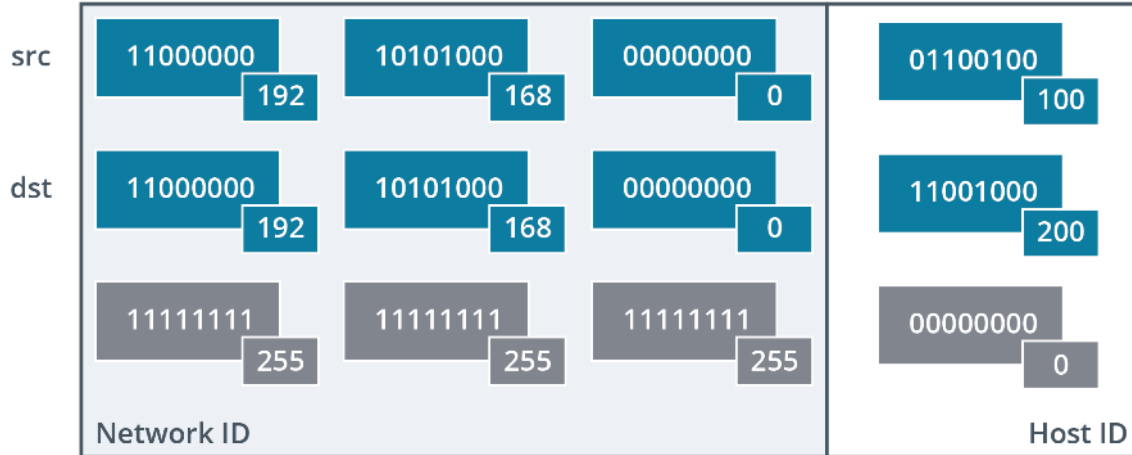
Network Prefixes

- Network ID-network the node belongs to
- Host ID- individual node
- Network mask- Subnet mask



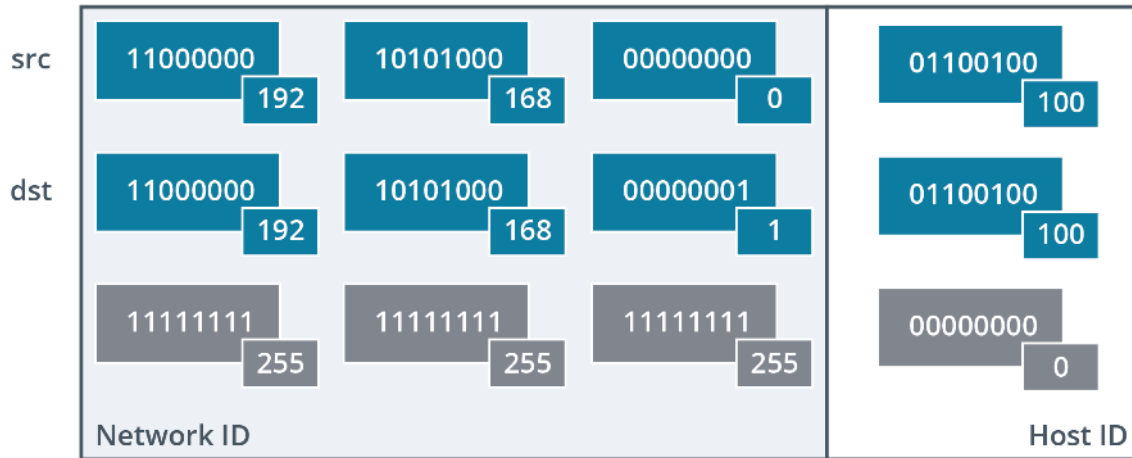
IPv4 Forwarding

- Same source and destination- forward back to same interface on router
- ARP table to identify local interface to forward packet to



IPv4 Forwarding

- Different Source and Destination Networks
 - Forward to appropriate interface on router



Public and Private Addressing

Private Ranges- Used within a LAN environment

- 10.0.0.0 to 10.255.255.255 (Class A)
- 172.16.0.0 to 172.31.255.255 (Class B)
- 192.168.0.0 to 192.168.255.255 (Class C)

Network Address Translation (NAT)

- Private to public
- Public to private

Public and Private Addressing

Class	1st Octet Range	Dotted Decimal Mask	Network Prefix	Binary Mask
A	0-127	255.0.0.0	/8	11111111 00000000 00000000 00000000
B	128-191	255.255.0.0	/16	11111111 11111111 00000000 00000000
C	192-223	255.255.255.0	/24	11111111 11111111 11111111 00000000

IPv4 Host Address Configuration

- IP assigned to node
- Subnet mask
- Default gateway
- DNS server IP

IPv4

☒ On

IP address

Subnet prefix length

Gateway

Preferred DNS

Alternative DNS

IPv6

SaveCancel

Static Versus Dynamic Host Address Configuration

Static- Manual configuration of NIC

DHCP- Service is used to configure NIC

APIPA- Automatic Private IP Addressing

- 169.254.0.1 to 169.254.255.254

Static Versus Dynamic Host Address Configuration

TP-LINK

Archer VR900

Quick Setup

Basic

Advanced

Logout

Reboot

Status

Operation Mode

Network

Internet

LAN Settings

Interface Grouping

DSL Settings

Dynamic DNS

Advanced Routing

IPSec VPN

IPTV

DHCP Server

IP Version:

☒ IPv4 ☐ IPv6

MAC Address:

60:E3:27:CF:EA:CB

LAN IPv4:

192.168.0.1

Subnet Mask:

255.255.255.0

IGMP Snooping:

☒ Enable IGMP Snooping

DHCP:

☒ Enable DHCP

☒ DHCP Server ☐ DHCP Relay

IP Address Pool:

192.168.0.100 - 192.168.1.199

Address Lease Time:

1440 minutes(1-2880)

Default Gateway:

192.168.0.1 (Optional)

Default Domain:

(Optional)

Primary DNS:

0.0.0.0 (Optional)

Secondary DNS:

0.0.0.0 (Optional)

Save

Screenshot courtesy of TP-Link.

Interfaces

- Multiple interfaces to connect various networks

Console vs. GUI/Web URL

SOHO Router Configuration

TP-LINK

Archer VR900

Quick SetupBasicAdvanced

LogoutReboot

Status

Operation Mode

Network

Internet

LAN Settings

Interface Grouping

DSL Settings

Dynamic DNS

Advanced Routing

IPSec VPN

IPTV

WAN Interface

RefreshAddDelete All

WAN Interface Name	VPI/VCI or VID	Status	Operation	Modify
pppoe_ptm_101_0_d	101	Connected	Disconnect	 

Internet Connection Setup

DSL Modulation Type:VDSL

VLAN ID:101

Internet Connection Type:PPPoE

Username:broadband.user@btbroa

Password:

Confirm password:

Connection Mode:

Always on

Connect on demand

Connect manually

Screenshot courtesy of TP-Link.

SOHO Router Configuration

TP-LINK
Archer VR900

Quick SetupBasicAdvanced

LogoutReboot

Status

Operation Mode

Network

IPTV

Wireless

Guest Network

NAT Forwarding

USB Settings

Parental Control

DSL

Line Status: Connected

DSL Modulation Type: VDSL2

Annex Type: Annex A/L/M

	Upstream	Downstream
Current Rate(kbps)	9999	35428
Max Rate(kbps)	19294	41045
SNR Margin(dB)	12.9	7.8
Line Attenuation(dB)	0	31.6
Errors(pkts)	0	0

Screenshot courtesy of TP-Link.

IPv6 Addressing

128-bit addressing;
hex notation

Eight 16-bit values
separated by colons

Prefixes

- Pre-determined address for special purposes

Dual stacking

- IPv4 and IPv6

IPv6 Addressing



Lab Activity

- Lab: Configure IP Addresses
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: TCP/IP Concepts

- TCP/IP
- Link or Network Interface Layer
- Internet Layer
- Transport Layer
- Application Layer
- IPv4 Addressing
- Network Prefixes
- IPv4 Forwarding
- Public and Private Addressing

Lesson Review: TCP/IP Concepts

- IPv4 Host Address Configuration
- Static Versus Dynamic Host Address Configuration
- SOHO Router Configuration
- IPv6 Addressing

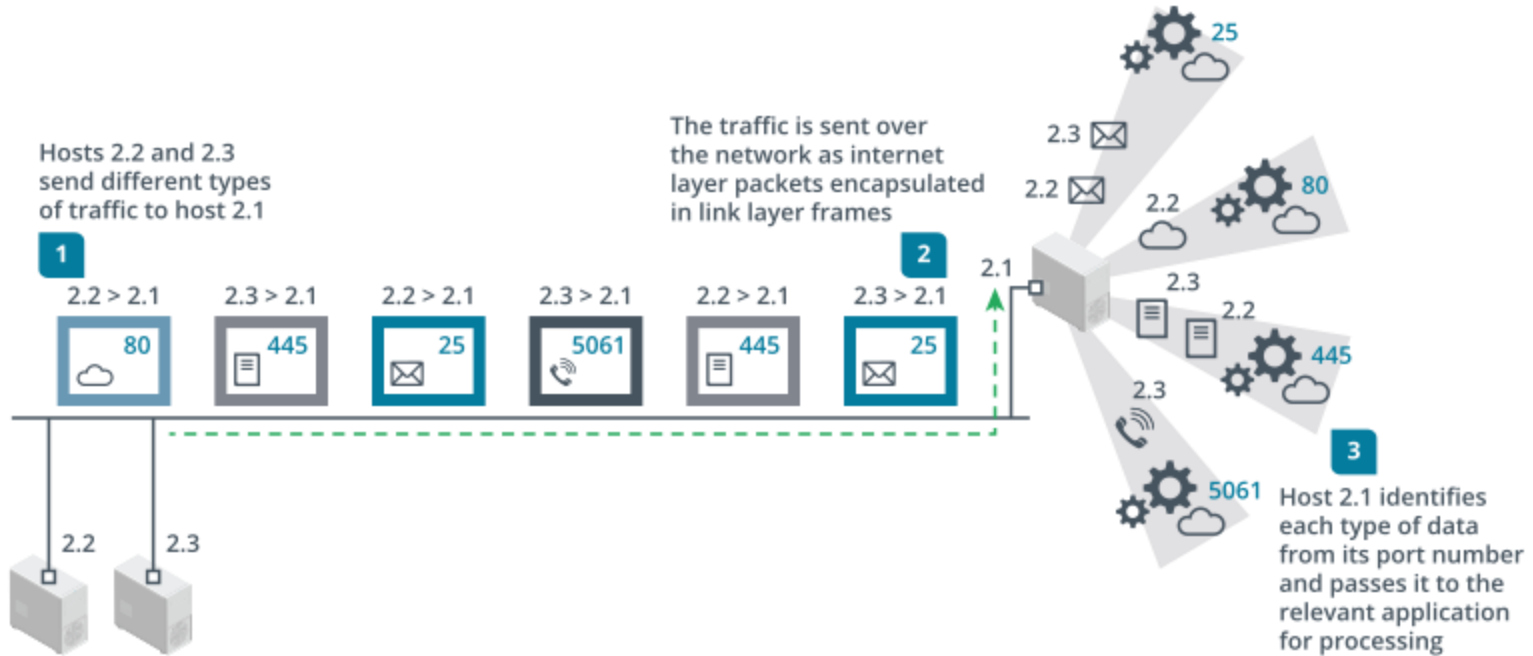
Lesson 6.3

Network Communications

Protocols and Ports

- Transport layer
- 65,536 ports (0-65535)
- Facilitate simultaneous communications

Protocols and Ports



Images © 123RF.com

Transmission Control Protocol

- Connection-oriented, guaranteed delivery
- Flags to control
- Three-way handshake
 - SYN - SYN/ACK - ACK
- Examples: HTTPS, SSH, FTP, etc.

Transmission Control Protocol

No.	Time	Source	Destination	Protocol	Length	Info
1	0.0...	10.1.0.101	10.1.0.2	TCP	66	1624 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
2	0.0...	10.1.0.2	10.1.0.101	TCP	66	80 → 1624 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=
3	0.0...	10.1.0.101	10.1.0.2	TCP	54	1624 → 80 [ACK] Seq=1 Ack=1 Win=262144 Len=0
4	0.0...	10.1.0.101	10.1.0.2	HTTP	433	GET / HTTP/1.1
5	0.0...	10.1.0.2	10.1.0.101	TCP	54	80 → 1624 [ACK] Seq=1 Ack=380 Win=2102272 Len=0
6	0.2...	10.1.0.2	10.1.0.101	TCP	1514	80 → 1624 [ACK] Seq=1 Ack=380 Win=2102272 Len=1460 [
7	0.2...	10.1.0.2	10.1.0.101	HTTP	770	HTTP/1.1 200 OK (text/html)

> Internet Protocol Version 4, Src: 10.1.0.101, Dst: 10.1.0.2

▼ Transmission Control Protocol, Src Port: 1624, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

Source Port: 1624

Destination Port: 80

[Stream index: 0]

[TCP Segment Len: 0]

Sequence number: 1 (relative sequence number)

[Next sequence number: 1 (relative sequence number)]

Acknowledgment number: 1 (relative ack number)

0101 = Header Length: 20 bytes (5)

> Flags: 0x010 (ACK)

Window size value: 1024

[Calculated window size: 262144]

[Window size scaling factor: 256]

0000	00 15 5d 01 ca 76 00 15 5d 01 ca 77 08 00 45 00	..].v..].w..E.
0010	00 28 16 49 40 00 80 06 00 00 0a 01 00 65 0a 01	.(.I@... ..e..
0020	00 02 06 58 00 50 61 ff 66 ce 99 26 04 92 50 10	...X.Pa. f..&..P.
0030	04 00 14 83 00 00	

Screenshot courtesy of Wireshark.

User Datagram Protocol



Connectionless, non-guaranteed delivery

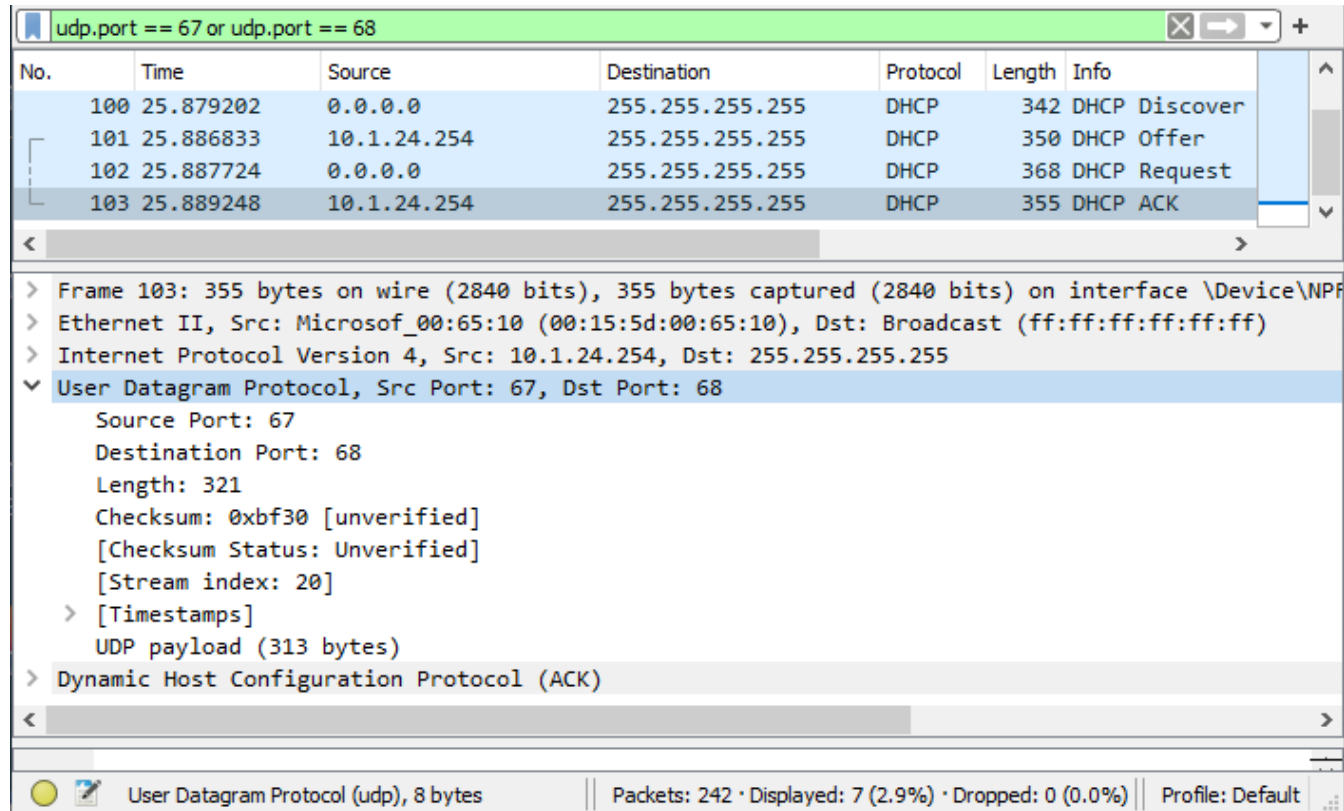


No sequencing or acknowledgments



Examples: DHCP, DNS, TFTP, etc.

User Datagram Protocol



The screenshot shows a Wireshark packet capture window with a filter set to `udp.port == 67 or udp.port == 68`. The packet list displays four packets related to a DHCP transaction. Packet 103 is selected, showing its details in the packet pane.

No.	Time	Source	Destination	Protocol	Length	Info
100	25.879202	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover
101	25.886833	10.1.24.254	255.255.255.255	DHCP	350	DHCP Offer
102	25.887724	0.0.0.0	255.255.255.255	DHCP	368	DHCP Request
103	25.889248	10.1.24.254	255.255.255.255	DHCP	355	DHCP ACK

Details of the selected packet (103):

- > Frame 103: 355 bytes on wire (2840 bits), 355 bytes captured (2840 bits) on interface \Device\NPF...
- > Ethernet II, Src: Microsof_00:65:10 (00:15:5d:00:65:10), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- > Internet Protocol Version 4, Src: 10.1.24.254, Dst: 255.255.255.255
- ▼ User Datagram Protocol, Src Port: 67, Dst Port: 68
 - Source Port: 67
 - Destination Port: 68
 - Length: 321
 - Checksum: 0xbf30 [unverified]
 - [Checksum Status: Unverified]
 - [Stream index: 20]
 - > [Timestamps]
 - UDP payload (313 bytes)
 - > Dynamic Host Configuration Protocol (ACK)

Summary bar: User Datagram Protocol (udp), 8 bytes | Packets: 242 · Displayed: 7 (2.9%) · Dropped: 0 (0.0%) | Profile: Default

Screenshot courtesy of Wireshark.

Well-Known Ports

- 20/21- FTP
- 22- SSH
- 23- Telnet
- 25- SMTP
- 53- DNS
- 67/68- DHCP
- 80- HTTP
- 110- POP3
- 137-139- NetBIOS
- 143- IMAP
- 389- LDAP
- 443- HTTPS
- 445- SMB
- 3389- RDP



Lesson Review: Network Communications

- Protocols and Ports
- Transmission Control Protocol
- User Datagram Protocol
- Well-Known Ports

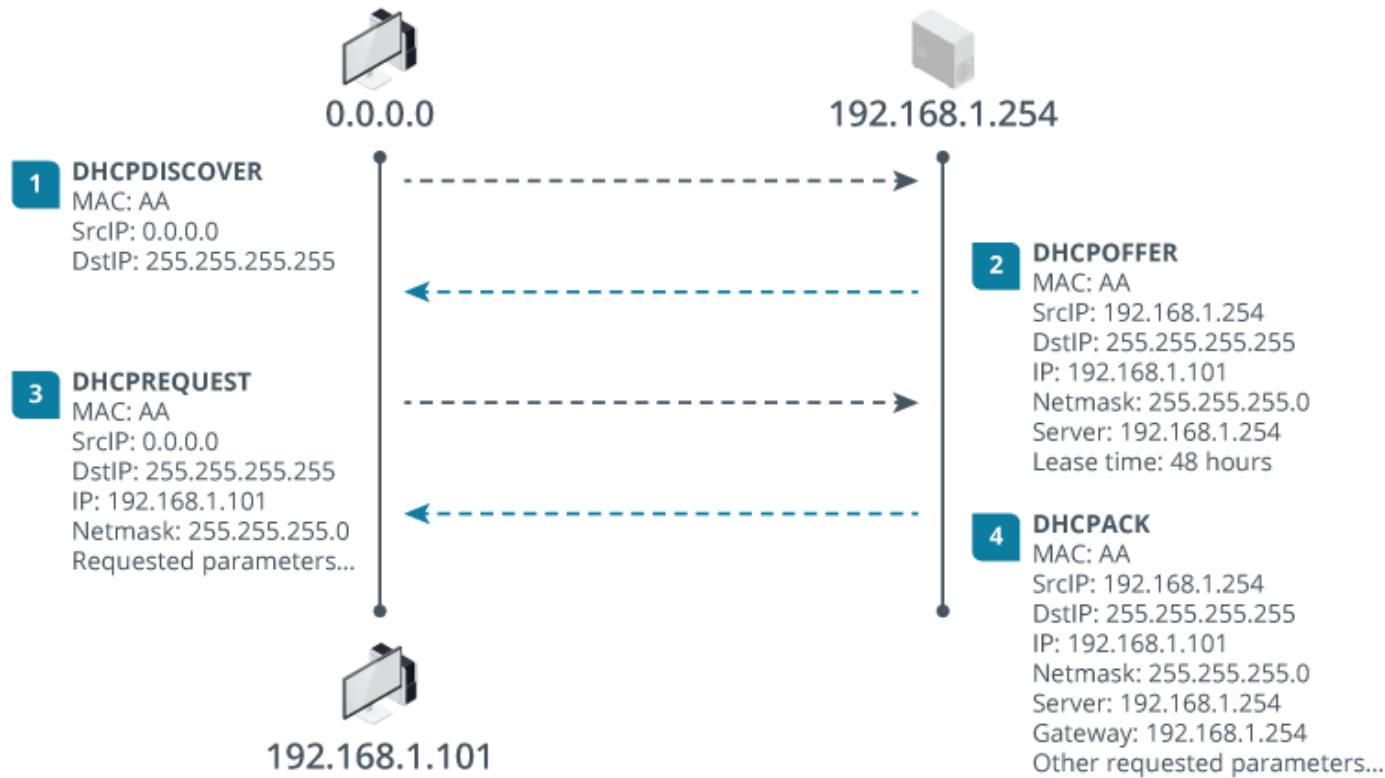
Lesson 6.4

Network Configuration Concepts

DHCP Functions

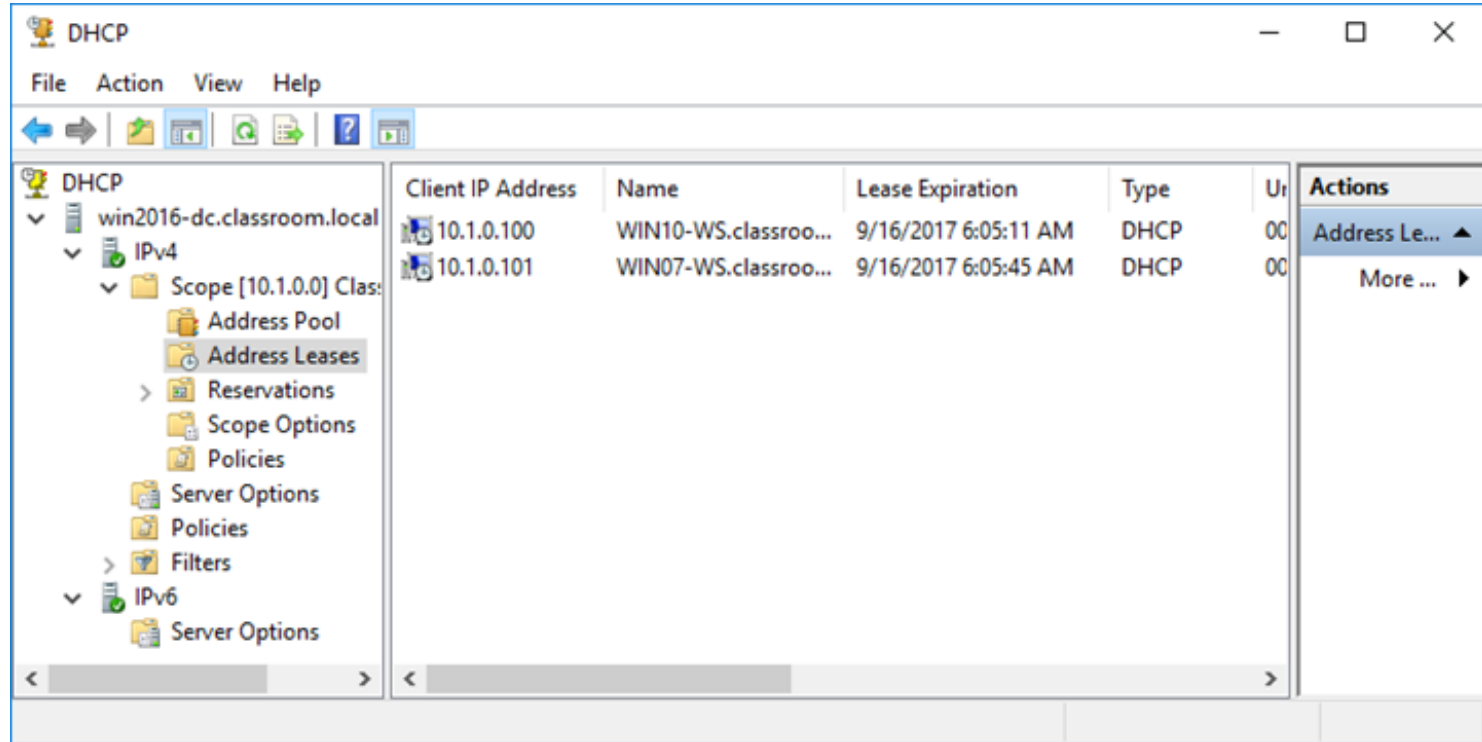
- Scope – The range of addresses to hand out
- Leases – The amount of time you will have the IP Address
- Reservations – Reserve addresses for devices that you want to always have the same address
 - MAC address needed
- DORA – Discover, Offer, Request and Accept

DHCP Functions



Images © 123RF.com.

DHCP Functions



Screenshot courtesy of Microsoft.

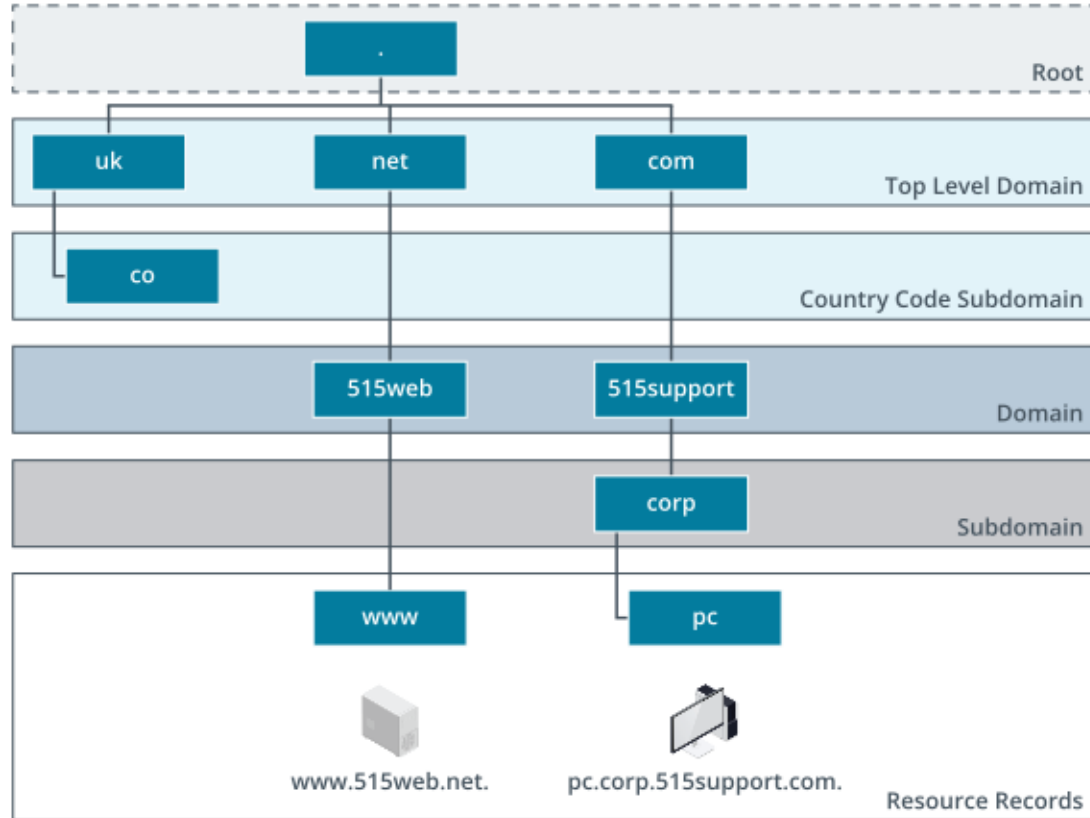
Domain Name System

DNS – makes the internet user friendly

FQDN or Host
Name to IP Address

TLD

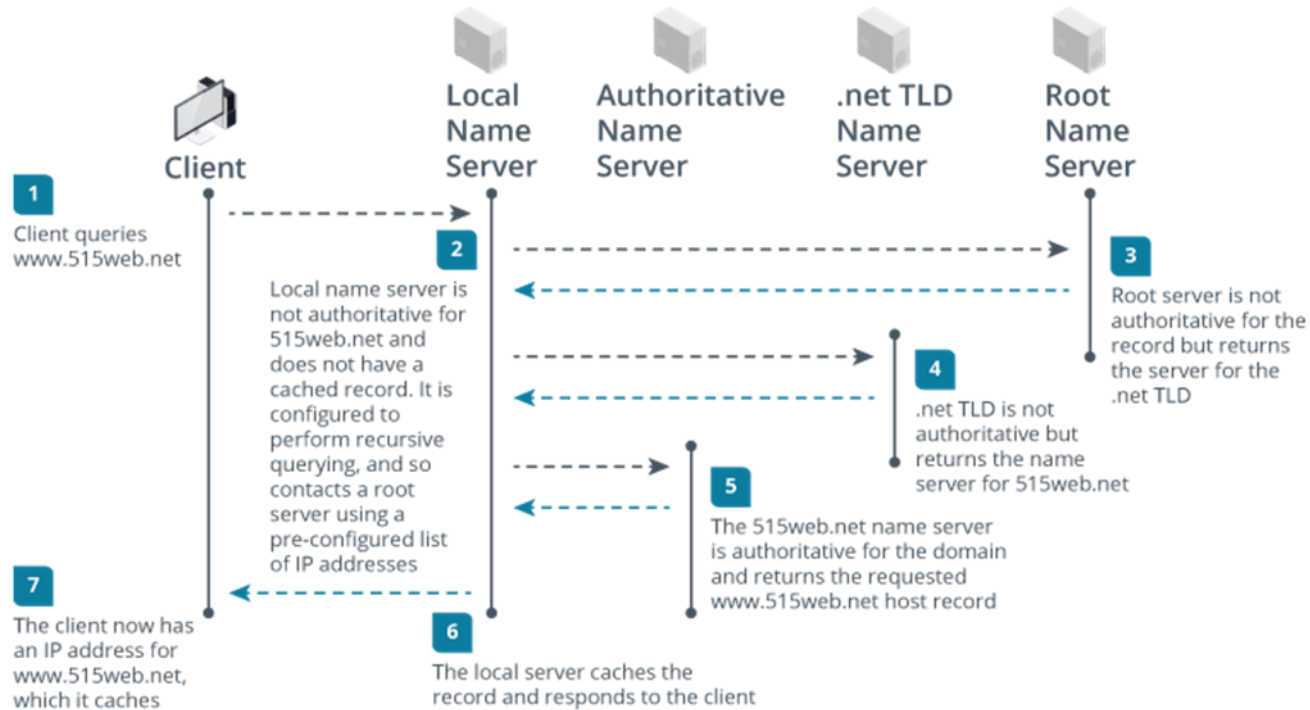
Domain Name System



DNS Queries

- Name resolution
 - HOSTS
 - Local server
 - Authoritative server
 - TLD server
 - Root name server

DNS Queries

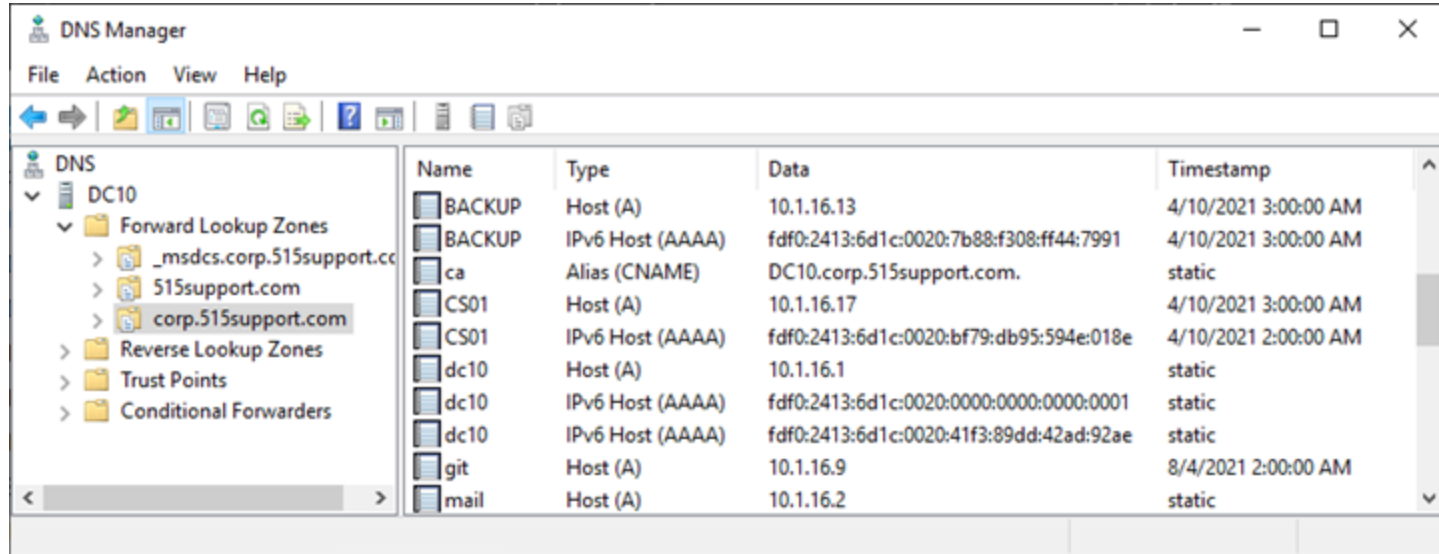


Images © 123RF.com

DNS Record Types

- **SOA** - (Start of Authority) record in DNS is a crucial record that defines the primary server for a domain
- **A** - DNS "A" record (Address record) is the most fundamental DNS record, mapping a human-readable domain name (like www.example.com) to its corresponding numerical **IPv4 address**
- **AAAA** - DNS "AAAA" record is the most fundamental DNS record, mapping a human-readable domain name (like www.example.com) to its corresponding numerical **IPv6 address**
- **CNAME** - (Canonical Name) record creates an **alias**, pointing one domain or subdomain (like www.example.com) to another "true" domain name (like example.com)
- **MX** - (Mail eXchange) record is a type of DNS record that tells the internet which mail servers are responsible for accepting emails for a specific domain
- **TXT** - (Text) record stores human-readable or machine-readable text about a domain
- **SRV** - (Service) record locates specific services on a network, providing the hostname, port, and protocol, not just an IP address
- **NS** - (Name Server) record specifies which DNS servers are authoritative for a domain, telling the internet where to find that domain's DNS information
- **PTR- Reverse A/AAAA** - (Pointer) record performs a **reverse DNS lookup**, mapping an **IP address back to a domain name**, which is the opposite of a standard A record (domain to IP)

DNS Record Types

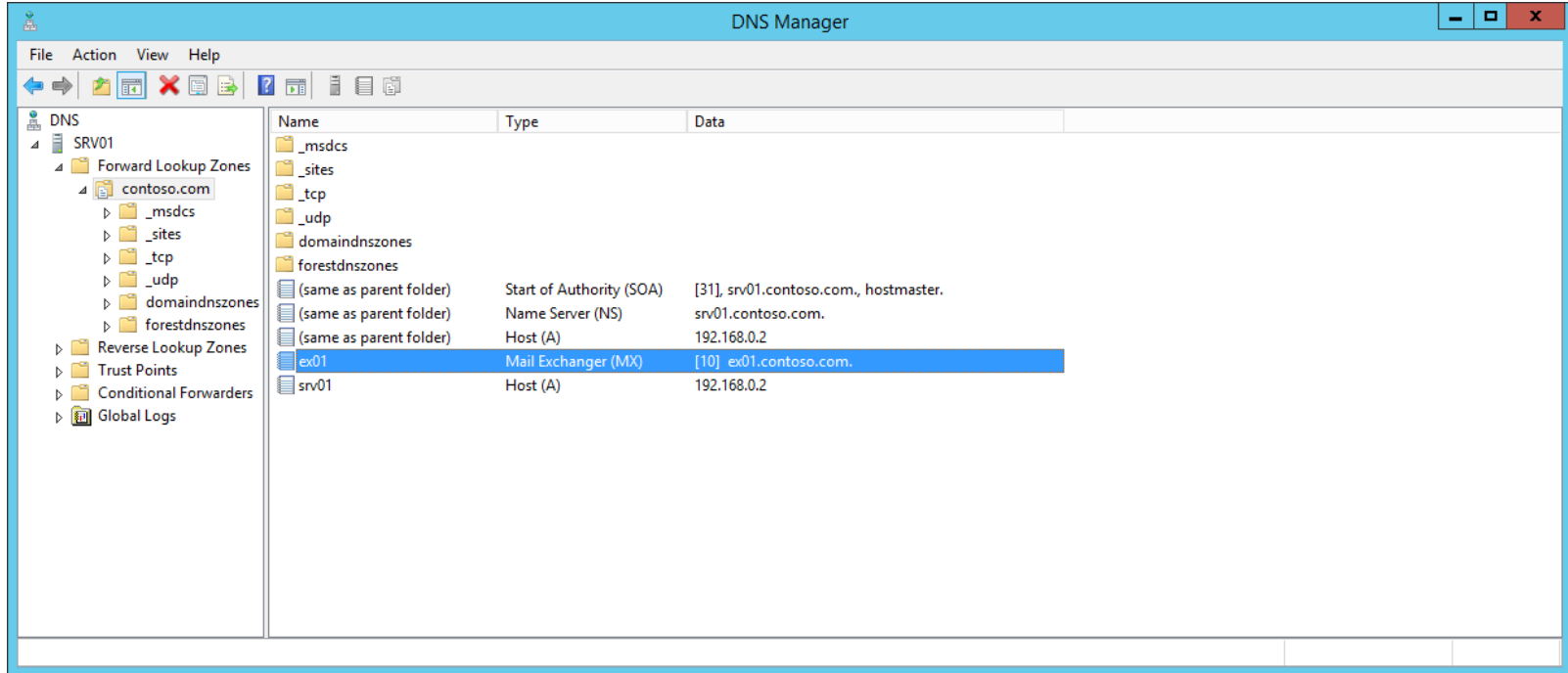


The screenshot shows the Windows DNS Manager application. The left pane displays the hierarchy: DNS > DC10 > Forward Lookup Zones > corp.515support.com. The right pane shows a table of DNS records for this zone.

Name	Type	Data	Timestamp
BACKUP	Host (A)	10.1.16.13	4/10/2021 3:00:00 AM
BACKUP	IPv6 Host (AAAA)	fdf0:2413:6d1c:0020:7b88:f308:ff44:7991	4/10/2021 3:00:00 AM
ca	Alias (CNAME)	DC10.corp.515support.com.	static
CS01	Host (A)	10.1.16.17	4/10/2021 3:00:00 AM
CS01	IPv6 Host (AAAA)	fdf0:2413:6d1c:0020:bf79:db95:594e:018e	4/10/2021 2:00:00 AM
dc10	Host (A)	10.1.16.1	static
dc10	IPv6 Host (AAAA)	fdf0:2413:6d1c:0020:0000:0000:0000:0001	static
dc10	IPv6 Host (AAAA)	fdf0:2413:6d1c:0020:41f3:89dd:42ad:92ae	static
git	Host (A)	10.1.16.9	8/4/2021 2:00:00 AM
mail	Host (A)	10.1.16.2	static

Screenshot courtesy of Microsoft.

DNS Record Types



Screenshot courtesy of Microsoft.

DNS Spam Management Records

- **Sender Policy Framework (SPF)** - an email authentication method that prevents spoofing and phishing by letting domain owners list authorized mail servers
- **DomainKeys Identified Mail (DKIM)** - an email authentication standard that uses cryptographic signatures to verify an email's sender and ensure it hasn't been altered in transit
- **DMARC** - (Domain-based Message Authentication, Reporting, and Conformance) uses DNS TXT records to tell receiving mail servers how to handle emails claiming to be from your domain that fail SPF or DKIM authentication

Virtual LANs

- Increased performance
 - Broadcast domain split
- Increased security
 - Control of communication between networks and nodes
- Implemented on switch

```
interface swp5
    bridge-access 100

interface swp6
    bridge-access 100

interface swp7
    bridge-access 100

interface swp8
    bridge-access 100

interface swp9
    bridge-access 200

interface swp10
    bridge-access 200

interface swp11
    bridge-access 200

interface swp12
    bridge-access 200

interface bridge
    bridge-ports swp5 swp6 swp7 swp8 swp9 swp10 swp11 swp12
    bridge-vids 10 100 200
    bridge-vlan-aware yes
```

Virtual Private Networks

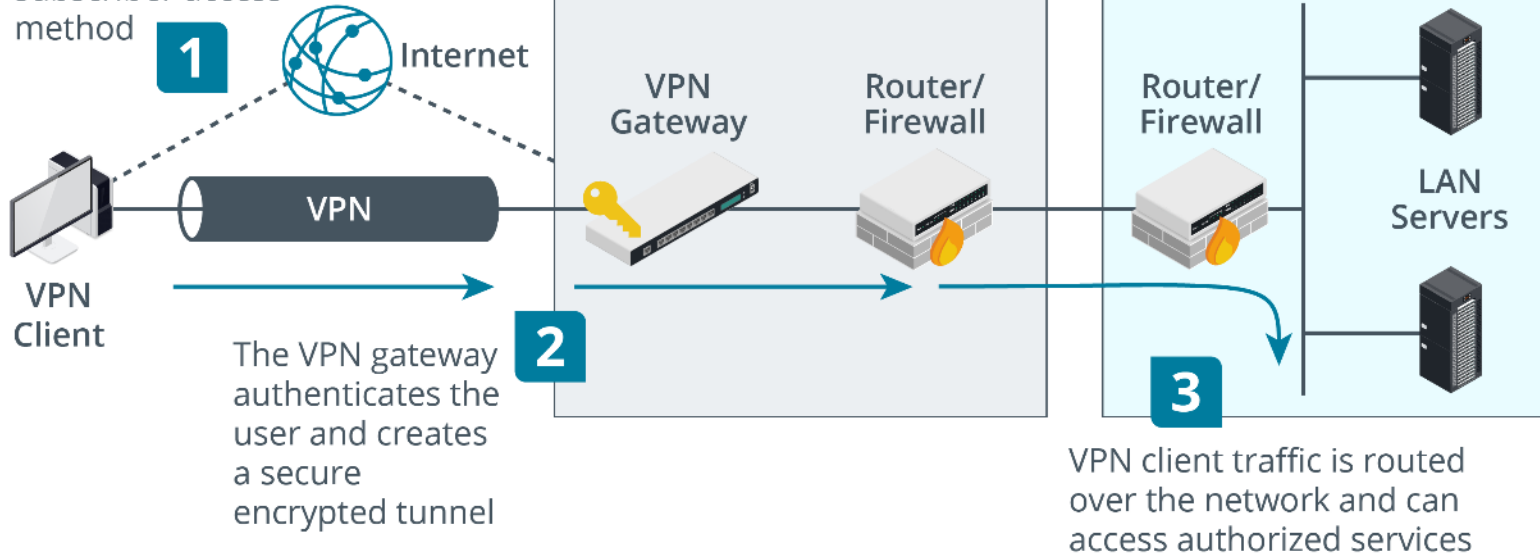
Remote connection method for security

Secure tunneled traffic over non-secure connections

- Public Wi-Fi

Virtual Private Networks

The VPN client host connects to a VPN gateway using any type of Internet subscriber access method



Images © 123RF.com

Lab Activity

- Challenge Lab: Install a SOHO Network
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Network Configuration Concepts

- DHCP Functions
- Domain Name System
- DNS Queries
- DNS Record Types
- DNS Spam Management Records
- Virtual LANs
- Virtual Private Networks